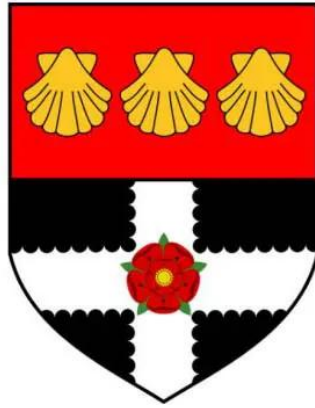




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ICM218 RESEARCH PROJECT

*Geopolitical Risk and Sectoral Volatility:
Evidence from US Energy and Defence
Sectors*

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The project is my own work, and the responsibility for errors is mine.

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Abstract

Geopolitical events transmit into financial markets through discontinuous, sector-wise jumps that conventional risk models fail to detect. This study constructs and tests a two-layer early warning system for the US Aerospace & Defence and Energy sectors, combining a dynamic realised-beta seismograph with a GARCH(1,1) volatility model and using the Geopolitical Risk Index of Caldara and Iacoviello (2022) as the underlying shock measure. The model is calibrated on the calm 2015 to 2019 period and tested on the out-of-sample period from 2020 to 2025, containing the COVID-19 pandemic, the Russian invasion of Ukraine, and the Hamas-Israel conflict. Results show that dynamic beta varies substantially more than the static OLS estimate. Main findings include the following: the seismograph detects regime shifts in real time, alarming at least a week before three of the five major events, GPR functions as a leading indicator of sector beta with a 10-trading-day predictive horizon and the usual convention that Aerospace & Defence being a safe-haven during crisis, fades after the Russian invasion of Ukraine and reverses sign in the post-Ukraine period. All findings are validated empirically by a combination of tests and checks throughout the study.

1. INTRODUCTION:

Geopolitical events are increasingly a crucial aspect of today's global markets. Their effects are asymmetric across sectors and poorly captured by traditional models. The 2022 Russian invasion of Ukraine is a clear example of this asymmetry: Brent crude crossed \$100 per barrel in a single session and the S&P 500 fell approximately 2%, but that aggregate move concealed a sharp surge in defence stocks, bidirectional movement in energy companies depending on Russian exposure, and a spike in the VIX (Rigobon and Sack, 2005; Kilian, 2009). These sudden changes are jumps rather than gradual updates, exactly the type of event that traditional risk models, built on assumptions of approximately normal return distributions and stable inter-asset relationships, systematically fail to anticipate (Bloom, 2009; Caldara and Iacoviello, 2022). Rigobon and Sack (2005), studying the 2003 Iraq war, showed that an identifiable war risk factor moved US Treasury yields and equity prices on a near-daily basis and could not be captured by conventional asset-pricing variables. Even today, the tools available to most portfolio managers have not been updated to integrate such nuances.

The major problem is that the techniques for measuring risk are backward-looking and linear. Beta, the measure of how much a sector moves in response to the overall market, is estimated through ordinary least squares regression and assumes that the sector-market relationship is approximately linear and stable over time (Sharpe, 1964). This assumption holds during calmer periods, but it averages calm and highly volatile windows together and therefore smooths out the exact moment when a portfolio manager needs to see a change the most (Fabozzi and Francis, 1978). Managers are forced to rely on tools that recognise damage after the shock has passed, leaving two important questions unanswered. The first is a timing question: at what point does the

current situation become different enough from the past that action is required? The second is a duration question: once high volatility prevails, how long is it likely to last? A mechanism that answers both, sector by sector, would change how institutional risk is managed during crises.

The recent series of geopolitical events has gained attention of researchers to study how they translate into asset prices (Caldara and Iacoviello, 2022; Gheorghe and Panazan, 2025). Among all sectors, Energy and Aerospace & Defence are the two most directly exposed, and their exposures are asymmetric in a way that warrants closer study. Defence has historically behaved as a geopolitical hedge: rising conflict expectations create higher demand for military equipment, so defence equities tend to perform strongly when the broader market falls (Apergis et al., 2017; Gheorghe and Panazan, 2025). Energy is more complex. Kilian (2009) showed that different classes of shocks - supply disruptions, aggregate demand changes, and precautionary shifts, all move oil prices in different directions, which is why the energy sector's response is bidirectional rather than one-directional. Observing both sectors is, in effect, a way of reading both sides of the same coin in the context of geopolitical events.

This research builds a two-layer early-warning framework designed to address the timing and duration problems identified above. The framework combines a non-parametric beta seismograph drawn from Hooper (2025), which estimates the relationship between each sector and the overall market using rolling windows of locally weighted regression (Cleveland, 1979), with a GARCH(1,1) model used to forecast how long the volatility regime triggered by the seismograph is likely to persist. When the estimated sector sensitivity diverges sharply from its rolling baseline, the seismograph alarms, signalling a regime shift as it occurs rather than after the fact. The

framework is calibrated on a calm period (2015 to 2019) and applied out-of-sample to a period of repeated crises (2020 to 2025) to test whether it detects regime shifts in real time and whether GPR carries information about sector beta beyond what is already captured by general market fear.

The study addresses three specific research questions, each tested empirically in the results section:

1. Can the seismograph detect regime shifts in real time with sufficient lead time to support portfolio action?
2. Whether GPR carries marginal predictive content for sector beta beyond VIX and market returns, and over what horizons?
3. Whether the GPR–beta relationship is uniform or regime-dependent across the 2020–2025 window?

This study makes three contributions. Empirically, it documents a regime-dependent GPR-beta relationship, a ten-trading-day predictive horizon from GPR to sector beta, and a structural change in the sign of the A&D coefficient following the Russian invasion of Ukraine. None of which has been previously documented in the GPR sector literature to the author's knowledge. Methodologically, it operationalises Hooper's (2025) conceptual seismograph framework, validates it out-of-sample across five major geopolitical events, and extends it with a GARCH-based duration forecast that turns the alarm into a two-part signal of both timing and persistence. In practice, the combination of a ten-day predictive lead time and a real-time seismograph alarm provides institutional portfolio managers with a structured early-warning system suitable for hedging timing decisions, sector rotation, and complementary risk

monitoring alongside conventional value-at-risk models.

2. LITERATURE REVIEW:

The literature on geopolitical risk in financial markets covers four overlapping concerns: how to measure something as qualitative as geopolitics, whether that measure actually moves asset prices, whether different sectors respond differently, and whether the standard tools we use to measure sector–market sensitivity still work in modern conditions. The biggest challenge in studying geopolitical risk empirically is that the concept is genuinely qualitative. The breakthrough came with Caldara and Iacoviello (2022), who built a daily Geopolitical Risk (GPR) index by scanning large newspaper archives for articles containing specific combinations of keywords linked to geopolitical threats and acts, such as war risk, military threat, and terrorist attack, then dividing the count by the total number of articles published that day. This turns an abstract concept into a continuous time series that econometric models can actually use. The GPR index is not a perfect measure, the keyword filter is English-language-biased, and the choice of newspapers reflects a Western editorial perspective. Despite these limitations, the daily frequency, long historical track record, and consistent construction methodology have made it the standard input for quantitative work on geopolitical risk, and it is the measure used throughout this study.

Once a continuous measure of GPR existed, a wave of empirical work tested whether it actually moves financial markets. Apergis, Bonato, Gupta, and Kyei (2017) applied non-parametric causality tests to a sample of major global defence companies and found that GPR predicts realised volatility in about half of them, although it does not predict returns in any consistent way. This return-versus-volatility distinction is more than a technicality. It is actually the empirical core of the field: geopolitical risk does

not tell you which direction prices will move, but it tells you when the magnitude of moves is about to increase. Bouras, Christou, Gupta, and Suleman (2019) extended this line of work to a panel of emerging stock markets using a Panel GARCH framework, confirming a positive link between GPR and market volatility, with country-specific geopolitical risk dominating the global GPR measure for most economies in their sample. Adekoya et al. (2025) extend this finding to the green-investments universe. In summary, while the GPR-volatility link and sectoral heterogeneity are well-documented, the existing literature lacks a temporal dimension and a systematic operationalisation of real-time detection tools. This study fills these gaps by validating Hooper's (2025) seismograph against the major 2020–2025 crises and identifying a ten-day predictive horizon from GPR to sector beta

The sectoral question matters in this study because the whole motivation for looking at Energy and Defence rests on the claim that these two sectors respond to geopolitical events in measurably different ways and are typically the first sectors to react. Gheorghe and Panazan (2025) provide the most direct evidence here. Using non-linear causality techniques across Defence, Cybersecurity, Energy, and Raw Materials, they show that defence consistently behaves as a geopolitical hedge, returns tend to rise when GPR rises, opposite to the broader market, while energy displays bidirectional exposure that depends on the type of shock. This bidirectionality is consistent with Kilian's (2009) much earlier decomposition of oil price shocks, which showed that supply disruptions, aggregate demand shifts, and precautionary demand changes move oil prices in different directions and at different speeds. A linear model that averages across these opposing reactions necessarily understates the sector's true sensitivity to geopolitical events. The findings of Gheorghe and Panazan therefore justify two decisions in this study: the choice of Energy and Defence as the two sectors of interest,

and the use of a non-linear estimation framework that can capture direction-dependent responses.

Most pre-2022 evidence treats defence stocks as a short-term hedge against transient geopolitical events. But SIPRI (2024) documents that the Ukraine war has produced the largest year-on-year increase in global military spending in decades, with NATO members committing to multi-year procurement programmes. The implication, is that the haven property of defence equities may not be stable across regimes. Sustained conflict environments may convert defence from a short-term hedge into a long-term procyclical exposure to geopolitical risk.

The methodological foundation comes from the time-varying beta literature. The static OLS beta from the standard market model (Sharpe, 1964) assumes a linear, constant sector-market relationship—an assumption that holds in calm periods but breaks down during crises (Fabozzi and Francis, 1978). Lewellen and Nagel (2006) refine this argument through the conditional CAPM framework, showing that beta is state-dependent and that averaging across states masks the true conditional sensitivity. A related strand documents how asset correlations shift during stress: Connolly, Stivers, and Sun (2005) show that the stock-bond correlation falls sharply during high-VIX periods, while Pástor and Veronesi (2013) develop a theoretical framework in which political uncertainty alters the correlation structure of stock returns. This study extends the decorrelation framework to the geopolitical context, testing whether sectoral equities act as partial havens under political uncertainty.

Hooper (2025) proposes a methodological framework that ties the time-varying beta and regime-detection ideas together. His non-parametric beta seismograph re-estimates beta over short rolling windows using locally weighted scatterplot smoothing

(LOWESS), a flexible regression technique formalised by Cleveland (1979), which allows the estimated relationship to bend smoothly in response to new data rather than being forced into a global linear form. When the estimated beta diverges sharply from its recent baseline, the seismograph alarms, flagging a potential regime shift in real time. Hooper's framework is, however, primarily conceptual. The paper proposes the tool but does not implement it on specific sectors, does not extend it with volatility-persistence forecasting, and does not validate it out-of-sample. This study does all three. By combining Hooper's seismograph with a GARCH model of volatility persistence, applying the combined system to the US Energy and Defence sectors, and testing it on a 2020–2025 out-of-sample period that contains three distinct geopolitical crises, this research moves the seismograph framework from concept to operational tool.

This study fills these gaps by operationalising and validating Hooper's (2025) seismograph across the major 2020–2025 crises, identifying a ten-day predictive horizon from GPR to sector beta, and documenting regime-dependent transmission across three distinct crisis episodes.

3. DATA:

This study uses daily-frequency data covering 2 January 2015 to 30 December 2025 (2,765 trading days). Daily frequency is chosen because the GPR index of Caldara and Iacoviello (2022) is published daily, and previous evidence shows that geopolitical shocks transmit into asset prices within hours rather than weeks (Rigobon and Sack, 2005). The global GPR measure is used, since US sectors are exposed to worldwide geopolitical events.

The sample is split into two sub-periods. The estimation period (2 January 2015 to 31 December 2019, 1,258 trading days) is a relatively calm window for US equities and is used to estimate the static OLS beta, the GARCH(1,1) parameters, the training-period correlation, and the seismograph thresholds. The out-of-sample testing period (2 January 2020 to 30 December 2025, 1,507 trading days) contains four distinct shocks: the COVID-19 pandemic, the Russian invasion of Ukraine, the European energy crisis, and the Hamas-Israel conflict. So, a model calibrated on calm data is being asked to perform during three of the most turbulent years in recent financial history.

Consistent with standard asset pricing models, total return indices are used to capture full investor returns, with the negligible daily risk-free rate omitted (Sharpe, 1964; Lewellen and Nagel, 2006).

Daily log returns are computed as $r(t) = \ln(P_t / P_{t-1})$, chosen for their time-additivity and superior statistical properties in GARCH estimation.

Variable	Description	Explanation
S&P 500 A&D Total Return Index	Market capitalisation weighted daily total return (price plus reinvested dividends) of all aerospace and defence companies in the S&P 500	Sector return series (Aerospace and Defence)
S&P 500 Energy Total Return Index	Market capitalisation weighted daily total return of all energy companies in the S&P 500	Sector return series (Energy)
S&P 500 Total Return Index	Market capitalisation weighted total return of 500 constituents	Market benchmark for beta estimation
CBOE Volatility Index (VIX)	30-day forward expected volatility of the S&P 500 implied by index option prices	Control variable for general market fear
GPR Index	Daily count of newspaper articles containing geopolitical threat and act keywords, scaled by total articles (Caldara and Iacoviello, 2022)	Independent variable: geopolitical shock measure
Dynamic Realised Beta $\beta(i,t)$	Daily sector–market sensitivity computed as $\rho \times \sigma_{\text{sector}(t)} / \sigma_{\text{market}(t)}$, where ρ is the training-period correlation and $\sigma(t)$ are GARCH(1,1) conditional volatilities	Dependent variable in the regression analysis

Table 3.1. Variables used in the study

The GPR index is used in two forms following standard practice (Bouras et al., 2019; Adekoya et al., 2025): the level, which captures the prevailing stock of geopolitical concern, and the daily absolute change, which isolates the shock component - the part of geopolitical risk that arrives unexpectedly. Both forms are constructed from the raw GPR index series without further transformation.

The dependent variable is the dynamic realised beta (Section 4.4). Two controls enter the regression: log VIX captures aggregate market fear, and the daily log return on the S&P 500 captures market direction. Including log VIX ensures the GPR coefficient captures only the marginal geopolitical signal, distinguishing sector-specific regime shifts from aggregate market panic.

4. METHODOLOGY:

4.1. Preliminary tests for data validity:

Before any modelling, two conditions need to hold: the data must be stationary, and the volatility must cluster. Both are standard requirements for the OLS regression and GARCH(1,1) estimation that follows.

The Augmented Dickey-Fuller test (Dickey and Fuller, 1979) is applied to the three log return series, which confirms stationarity of the three log return series. The GPR index is used in levels and daily absolute changes, following Bouras et al. (2019) and Adekoya et al. (2025).

An ARCH-LM test (Engle, 1982) is applied to the squared residuals of each sector return series, with the Ljung–Box Q-statistic on squared residuals (Ljung and Box, 1978) to confirm volatility clustering and justify the use of GARCH(1,1) .

4.2. Static OLS beta as a baseline:

To establish what normal sector sensitivity looks like, a static beta is estimated for each sector over the full estimation period using a standard OLS regression:

$$r(i,t) = \alpha(i) + \beta(i) \cdot r(m,t) + \varepsilon(i,t)$$

Here, $r(i,t)$ is the log return on sector i , $r(m,t)$ is the log return on the S&P 500, $\beta(i)$ is the sector's market beta, and $\varepsilon(i,t)$ is the residual. The risk-free rate is left out because at daily frequency the Treasury bill rate is negligible relative to equity returns and does not meaningfully affect beta estimates (Lewellen and Nagel, 2006). This is the standard market model formulation of Sharpe (1964): a beta > 1 means the sector amplifies as the market moves, while a beta below one means it is more defensive. The well-known limitation, recognised in the time-varying beta literature (Fabozzi and Francis, 1978; Lewellen and Nagel, 2006), is that this estimate assumes the sector–market relationship stays constant, but during a geopolitical crisis, the true beta can double or triple within weeks.

4.3. GARCH(1,1) modelling of conditional volatility:

A parsimonious GARCH(1,1) specification is used to capture the volatility clustering and persistence that characterise daily equity returns (Engle, 1982; Bollerslev, 1986; Hansen and Lunde, 2005). The model takes the form:

$$r(i,t) = \mu(i) + \varepsilon(i,t), \varepsilon(i,t) = \sigma(i,t) \cdot z(i,t), z(i,t) \sim N(0,1) \sigma^2(i,t) = \omega + \alpha \cdot \varepsilon^2(i,t-1) + \beta \cdot \sigma^2(i,t-1)$$

Where the sum $\alpha + \beta$ measures persistence. Values close to one indicate elevated volatility regimes that decay over many days rather than diminish overnight. GARCH parameters are estimated on the training period to characterise normal volatility, then re-estimated on the full sample to produce the conditional volatility series $\sigma(i,t)$ and $\sigma(m,t)$ used to build dynamic beta in Section 4.4.

4.4. Dynamic realised beta and the seismograph:

The dynamic realised beta produces a new beta for every trading day, following Hooper (2025):

$$\beta(i,t) = \rho(i) \cdot \sigma(i,t) / \sigma(m,t)$$

where $\rho(i)$ is the unconditional correlation between sector i 's returns and market returns over the training period, and $\sigma(i,t)$ and $\sigma(m,t)$ are the GARCH-estimated conditional volatilities of the sector and the market on day t . The correlation $\rho(i)$ is held fixed at its training-period value rather than estimated dynamically. Fixing ρ lets all daily variation in beta come from GARCH-driven volatility components, keeping the model parsimonious and the source of beta variation transparent (Bollerslev, Engle and Wooldridge, 1988). When sector volatility rises faster than market volatility, beta rises; when the two move together, beta stays close to its long-run value. As GARCH updates the volatilities each day, the beta updates with it.

The GARCH persistence then provides the duration signal. Because $\alpha + \beta$ is close to one for both sectors, once an elevated volatility regime starts, it tends to decay over many days rather than disappear overnight. A portfolio manager observing an alarm therefore gets two complementary pieces of information at once: when the regime has shifted, and roughly how long the elevated state is likely to last. Together, these two signals form the two-layer early-warning system.

4.5. Pre-crisis stress testing and scenario analysis:

Before applying the model out of sample, its behaviour is checked under controlled scenarios to see whether the predicted beta responds to GPR in a way consistent with

the existing literature. Predicted dynamic beta is computed at three representative GPR levels - normal (GPR = 100), high-tension (GPR = 250), and conflict (GPR = 400). The predictions come from a LOWESS fit (Cleveland, 1979) of dynamic beta on GPR using the full sample rather than only the estimation window, so that observations at the lower end of the GPR range are dense enough for the smoother to estimate the relationship reliably. This follows the compass methodology of Hooper (2025), where the scenario surface is mapped using all available data, since the purpose is structural description rather than predictive testing. Based on Apergis et al. (2017) and Gheorghe and Panazan (2025), the expectation is that both sectors will show elevated beta at higher GPR levels if they amplify systematic risk, or reduced beta if sector-specific drivers dominate.

4.6. Out-of-sample testing and multiple regression:

The model is applied to the 2020 to 2025 testing period without further adjustment. Seismograph alarms are checked against a timeline of major events, namely, the COVID-19 pandemic declaration (March 2020), the Russian invasion of Ukraine (February 2022), the Nord Stream pipeline sabotage (September 2022), and the Hamas-Israel attack (October 2023). GARCH volatility forecasts are compared to a 21-day rolling realised volatility using Mean Absolute Error and Root Mean Squared Error, following standard practice (Andersen and Bollerslev, 1998).

To statistically confirm the GPR–beta relationship while controlling for general market conditions, a multiple regression is estimated:

$$\beta(i,t) = \gamma_0 + \gamma_1 \cdot GPR(t) + \gamma_2 \cdot \log VIX(t) + \gamma_3 \cdot \log r(m,t) + u(i,t)$$

where $\beta(i,t)$ is the dynamic realised beta, $GPR(t)$ is the daily GPR index, $\log VIX(t)$ is

the natural log of the VIX, and $\log r(m,t)$ is the daily log return on the S&P 500. The GPR coefficient γ_1 is expected to be positive based on the existing literature (Apergis et al., 2017; Gheorghe and Panazan, 2025). The VIX coefficient γ_2 is expected to be positive, since higher market fear lifts sector betas since the VIX captures aggregate market-wide volatility expectations, and higher general fear typically increases co-movement between sectors and the broader market through systemic risk repricing. The market return coefficient γ_3 is sign-ambiguous since the relationship between contemporaneous market returns and dynamic beta is conditional on the regime: positive during stressed days when sectors amplify market declines, and negative or insignificant during calmer windows when sectors decouple from directional market moves. The overall specification follows the broader GPR–volatility literature (Bouras et al., 2019; Adekoya et al., 2025).

Including $\log VIX$ and log market returns as controls is what allows γ_1 to be interpreted as the marginal effect of GPR holding market fear and direction constant. Without these controls, the GPR coefficient would absorb co-movement with VIX, which is itself sensitive to geopolitical events, and blur the geopolitical signal with general directional pressure on prices. The log transformation linearises the relationship of VIX and market returns with beta and reduces the influence of extreme values, consistent with Bouras et al. (2019).

4.7. Additional empirical specifications:

Additional specifications test robustness, predictive horizon, and regime-dependence.

The baseline regression uses GPR in level form. An alternative specification re-estimates the regression with the daily absolute change in GPR as the independent

variable, following Bouras et al. (2019). This isolates the shock component of the geopolitical risk, specifically the unexpected daily movement rather than the overall level.

To test whether GPR contains information about future beta movements instead of just contemporaneous ones, the baseline regression is re-estimated with GPR entered at four horizons: lag 0 (same day), lag 1, lag 5, and lag 10 trading days. The OOS window covers three distinct crisis episodes that differ in nature and duration. To check whether the GPR–beta relationship is uniform across the OOS period or driven mainly by one episode, the baseline regression is re-estimated separately on three subsamples: the COVID era (2020–2021), the Ukraine and energy crisis era (2022), and the post-Ukraine era (2023–2025). This identifies regime-specific patterns that would otherwise be averaged out in the full OOS regression.

A further check tests whether the effect of GPR on beta changes depending on the prevailing VIX regime. An interaction term ($\text{GPR} \times \text{VIX}$) is added to the baseline regression. A significant interaction would mean GPR transmits differently in high-fear versus low-fear environments. GARCH-implied conditional volatility is compared against 21-day rolling realised volatility using MAE and RMSE (Andersen and Bollerslev, 1998), and a variance ratio F-test compares dynamic beta variance between training and out-of-sample periods to confirm structural regime change.

5. RESULTS AND DISCUSSION:

Diagnostic tests confirm stationarity (ADF $p < 0.01$ for all return series) and volatility clustering (ARCH-LM $p < 0.001$ for both sectors), validating the GARCH(1,1) specification (Table 5.1).

Test	Series	p-value	Result
ADF (stationarity)	S&P 500 log returns	0	Stationary
ADF	A&D log returns	0	Stationary
ADF	Energy log returns	0	Stationary
ARCH-LM (clustering)	A&D residuals	0	ARCH effects present
ARCH-LM	Energy residuals	0	ARCH effects present

Table 5.1: Preliminary Test Results

5.1. Static OLS beta and GARCH(1,1) estimation:

Estimated over the 2015 to 2019 training period, the static OLS beta is 1.0946 for Energy ($t = 34.75$, $R^2 = 0.49$) and 0.9267 for A&D ($t = 42.72$, $R^2 = 0.59$). Both regressions are statistically significant at the 1% level, indicating that during the calm window, Energy was slightly more sensitive to the market than A&D.

The GARCH(1,1) estimation produces highly persistent volatility series for both sectors and the market (Table 5.2). The market itself reacts most sharply to new shocks ($\alpha = 0.2087$), more than twice as high as either sector (Energy $\alpha = 0.0874$; A&D $\alpha = 0.0684$). Once volatility takes hold, though, the sectors retain it longer: A&D persistence is 0.9794, and Energy persistence is 0.9717, both above the market's 0.9470. This pattern is consistent with the practical reality that defence procurement contracts and energy supply agreements take months rather than days to settle, so shocks to these sectors transmit into multi-week volatility regimes.

Series	ω (omega)	α (alpha)	β (beta)	Persistence ($\alpha + \beta$)
S&P 500	0.0428	0.2087	0.7383	0.947
Energy	0.0481	0.0874	0.8843	0.9717
A&D	0.0218	0.0684	0.911	0.9794

Table 5.2: GARCH(1,1) Results

5.2. Dynamic realised beta:

The dynamic realised beta varies far more than the static OLS estimate. Over the calm 2015 to 2019 period, Energy's dynamic beta has a mean of 1.2187 and a maximum of 2.3939, while A&D's dynamic beta has a mean of 1.0343 and a maximum of 1.7851, illustrating that even in a relatively calm window, the sector–market relationship is not constant.

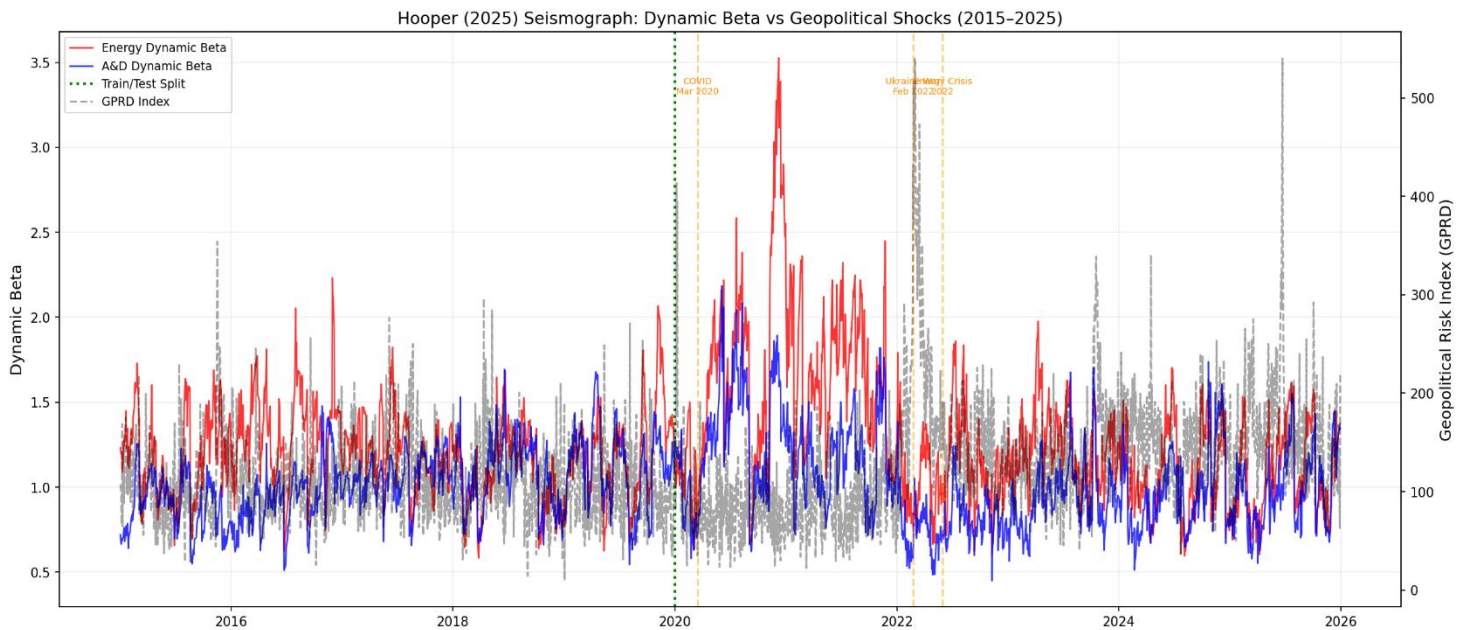


Figure 5.1: Hooper 2025 Seismograph

In the out-of-sample 2020 to 2025 testing period, both magnitudes and dispersion increase materially. The Energy beta has a mean of 1.3032 and reaches a high of 3.5274 on 8 December 2020, while the A&D beta reaches a peak of 2.1846 on 4 June 2020 (Figure 5.1). Both peaks occur during the COVID-19 pandemic, not during the highest

GPR readings of the sample.

This pattern raises an obvious question: if the seismograph is meant to detect geopolitical regime shifts, why are its largest detected shifts pandemic-driven? Both peaks occur during COVID rather than at the highest GPR readings, because the dynamic beta formula $\beta = \rho \times \sigma_{\text{sector}} / \sigma_{\text{market}}$ is mechanically most responsive to large sector-versus-market volatility differentials, and COVID produced the largest such differential in the sample. The geopolitical contribution to beta dynamics emerges across the OOS period as elevated regression coefficients rather than as discrete peaks. The seismograph successfully detects regime shifts that the static OLS beta could never have signalled, but the source of those shifts is correctly diagnosed by the regression evidence rather than read off the peak dates alone.

5.3. Out-of-sample multiple regression:

The training-period regression showed that GPR was not a significant predictor of dynamic beta once VIX and market returns were included as controls. Re-estimating the same specification on the out-of-sample window (2020–2025) produces a different result. The GPR coefficient is now highly significant and negative in both sectors: Energy at -0.0025 ($t = -14.79$, $p < 0.001$) and A&D at -0.0012 ($t = -10.86$, $p < 0.001$). Explanatory power also rises

Variable	Train Coef	Train t	Train p	OOS Coef	OOS t	OOS p
<i>Panel A: Energy</i>						
GPRD Index	0	0.19	0.846	−0.0025***	−14.79	<0.001
Log VIX	1.0161***	6.16	<0.001	1.0109***	5.21	<0.001
Log Total Returns	7.0251***	4.4	<0.001	5.2120***	4.48	<0.001
R ² / Adj R ²	0.0303 / 0.0280	-	-	0.1442 / 0.1425	-	-
<i>Panel B: Aerospace & Defence sector</i>						
GPRD Index	−0.0001	−0.88	0.379	−0.0012***	−10.86	<0.001
Log VIX	0.8161***	5.88	<0.001	0.7218***	5.56	<0.001
Log Total Returns	6.2963***	4.69	<0.001	3.5998***	4.62	<0.001
R ² / Adj R ²	0.0282 / 0.0259	-	-	0.0932 / 0.0914	-	-

Table 5.3: Multiple regression of dynamic beta on GPR, VIX, and market returns

substantially, from $R^2 = 0.03$ in the training period to 0.14 for Energy and 0.09 for A&D out-of-sample.

The dominance of the VIX coefficient confirms its role as the market's primary fear gauge, but the GPR seismograph functions as a specialist early-warning layer that captures the ten-day predictive horizon (Section 5.5). During the calm 2015-2019 window, GPR levels were low and relatively stable, so the small variation observed was already captured by VIX, leaving no marginal effect for GPR to pick up. This finding addresses the second research question and is consistent with the regime-conditional patterns documented by Apergis et al. (2017), Bouras et al. (2019), and Bloom (2009). The negative sign of the coefficient, however, contradicts the conventional expectation that risk rises with geopolitical stress. The modest R^2 of 0.14 reflects that beta dynamics have many drivers beyond GPR, VIX, and market returns; the regression is designed to test for the presence of a regime-specific effect, not to fully explain beta variation.

5.4. Sectoral decorrelation under stress:

To gauge the magnitude of the negative GPR coefficient, predicted dynamic beta is computed at three representative GPR levels using the LOWESS fit across the full sample. As reported in Table 5.4, beta declines monotonically as GPR rises in both sectors. At GPR = 400 (conflict regime), Energy beta is approximately 17% lower, and A&D beta is approximately 14% lower than at GPR = 100 (normal regime).

Scenario	Energy beta	A&D beta	Energy growth	A&D growth
Normal (GPR = 100)	1.2255	1.0086	1.000x	1.000x
High Tension (GPR = 250)	1.1265	0.8995	0.919x	0.892x
Conflict / Shock (GPR = 400)	1.0205	0.8701	0.833x	0.863x

Notes: Full sample period (2 Jan 2015 – 30 Dec 2025)

Table 5.4: Scenario Analysis Results

Figure 5.2 plots dynamic beta against the GPR index for both sectors with a LOWESS curve overlaid. The downward slope is visible across the full GPR range and holds for both Energy and A&D, confirming the scenario-table result graphically and consistent with the sectoral asymmetry that Gheorghe and Panazan (2025) document under nonlinear causality methods.

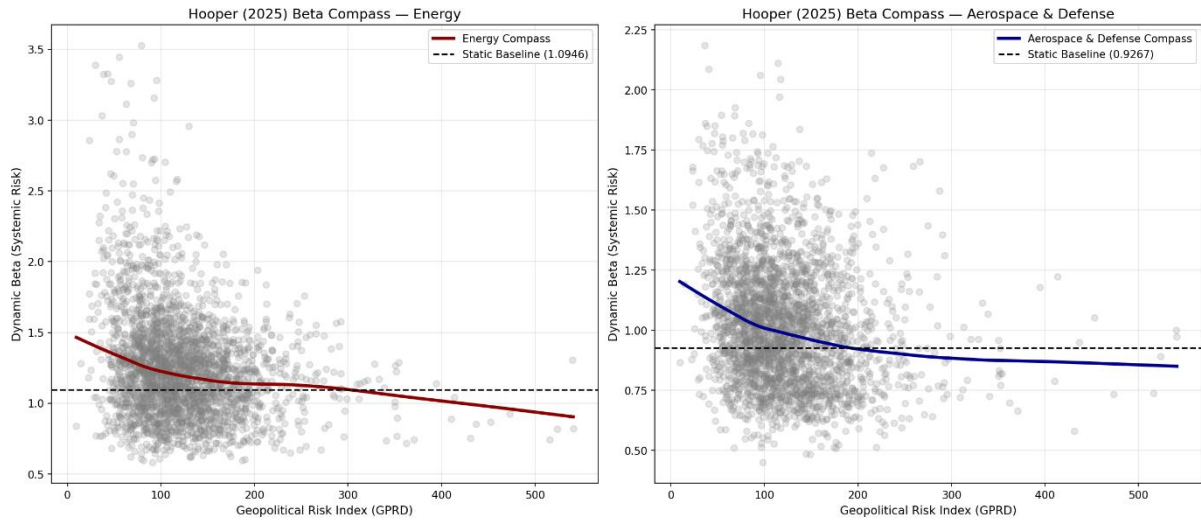


Figure 5.2: LOWESS Curves for Energy and A&D sectors

The mechanism here is decorrelation, not de-risking. Dynamic beta is defined as $\rho \times \sigma_{\text{sector}} / \sigma_{\text{market}}$ (Hooper, 2025), so a fall in beta can come either from a decline in the sector-market correlation ρ or from a relative drop in sector volatility. This distinction matters: a fall in beta from decorrelation does not imply that the sectors have become less risky in absolute terms as their underlying volatility may remain unchanged or rise, only the fact that their movement has become less synchronised with the broader market. Sector-specific drivers like oil supply for Energy (Kilian, 2009), defence procurement for A&D (Apergis et al., 2017) are not synchronised with the broader market, so the correlation falls while the volatility ratio remains stable. This flight-to-quality channel is consistent with the equity-bond decorrelation documented by Connolly, Stivers and Sun (2005) under high-VIX conditions and extending Pástor and Veronesi (2013) on political uncertainty and stock correlations to the geopolitical context. The interpretation also fits the conditional CAPM logic of Lewellen and Nagel (2006).

To check that this pattern is structural rather than a GPR-specific artefact. The same downward shape emerges in both sectors, indicating that regime dependence is a feature of how these sectors respond to stress generally (Figure 5.3).

The level coefficient reported above is negative, but the absolute-change coefficient in the Section 4.7.1 specification is positive. Rather than treating these as contradictory, they reflect two economically distinct transmission channels operating at different frequencies. The flight-to-quality channel (above) describes the structural relationship between *prevailing* GPR levels and *average* beta over weeks. On individual days when GPR rises sharply - invasions, attacks, major escalations - both sectors and the broader market spike together as investors price in the new information across all equities at once. This is the panic transmission channel, consistent with Bloom's (2009) finding that uncertainty shocks produce sharp, temporary contractions across economic and financial activity simultaneously. A study restricted to GPR levels would miss the shock channel; a study restricted to GPR changes would miss the level channel.

5.5. Lagged GPR:

To test whether GPR contains information about future beta movements rather than only contemporaneous ones, the OOS regression of Section 5.3 is re-estimated with GPR entered at four horizons: lag 0, lag 1, lag 5, and lag 10 trading days. Results are reported in Table 5.5. All four lags are highly significant ($p < 0.01$) in the OOS period, whereas training-period results are at most marginally significant and inconsistent in sign. The predictive content of GPR builds over time rather than fading: the lag-10 coefficient (Energy: -0.0011) carries the strongest signal, indicating that GPR levels observed two trading weeks earlier explain dynamic beta meaningfully and arguably more cleanly than same-day GPR, which is partly contaminated by simultaneous market reactions.

The implication is that GPR functions as a leading indicator rather than a simultaneous one. This is consistent with Caldara and Iacoviello (2022), whose impulse response analysis shows that GPR shocks transmit into stock returns and economic activity over a window of several days to weeks rather than instantaneously, and with Bloom (2009). From an applied perspective, a ten-day predictive horizon supports the seismograph framework of Hooper (2025) as a forecasting tool rather than a purely descriptive one: a portfolio manager observing elevated GPR today has approximately two trading weeks before the full effect on sector beta is realised, which is a meaningful operational window for hedging or rebalancing.

Sector	Lag	Training Period			Out-of-Sample Period		
		Coef	t-stat	p-value	Coef	t-stat	p-value
Energy	Lag 0	−0.0001	−0.40	0.686	−0.0010***	−3.93	<0.001
	Lag 1	0.0001	0.6	0.549	−0.0008***	−3.09	0.002
	Lag 5	0.0003*	1.73	0.084	−0.0008***	−3.68	<0.001
	Lag 10	0	0.18	0.858	−0.0011*** *	−5.51	<0.001
A&D	Lag 0	−0.0001	−0.63	0.526	−0.0004**	−2.15	0.031
	Lag 1	0	−0.02	0.982	−0.0003**	−2.05	0.04
	Lag 5	0.0001	0.38	0.704	−0.0005***	−3.27	0.001
	Lag 10	−0.0003	−1.65	0.099*	−0.0009*** *	−6.27	<0.001

Training period: 1,258 observations. Out-of-sample period: 1,507 observations. p-values below 0.001

Table 5.5: Lagged GPR regression

5.6. Subsample regressions: regime-dependent transmission:

The OOS sample combines three distinct crisis episodes that differ in nature, duration, and sectoral consequences. To examine whether the GPR–beta relationship documented in Section 5.3 is uniform across the OOS period or driven primarily by a particular regime, the regression

is re-estimated separately on three subsamples: COVID (2020–2021), Ukraine and the energy crisis (2022), and post-Ukraine (2023–2025). Results are reported in Table 5.6.

Regime	Variable	Sector					
		Energy sector			A&D sector		
		Coef	t-stat	p-value	Coef	t-stat	p-value
COVID era (2020–2021)							
	GPR	−0.0029***	−5.22	<0.001	−0.0009***	−2.60	0.009
	Log VIX	1.2809***	—	<0.001	0.7231***	—	<0.001
	Mkt Returns	5.1199***	—	0.007	3.3880***	—	0.003
	R ² / N	0.0747		505	0.0365		505
Ukraine era (2022)							
	GPR	−0.0006***	−3.21	0.002	0	−0.40	0.692
	Log VIX	0.8043**	—	0.035	0.4708**	—	0.03
	Mkt Returns	4.2579**	—	0.011	2.0858**	—	0.028
	R ² / N	0.0646		251	0.0224		251
Post-Ukraine era (2023–2025)							
	GPR	0	−0.17	0.862	0.0003*	1.93	0.054
	Log VIX	0.9104***	—	<0.001	0.6965***	—	<0.001
	Mkt Returns	5.8223***	—	<0.001	3.0164**	—	0.035
	R ² / N	0.0253		751	0.028		751

Table 5.6: Subsample regressions by crisis regime

5.6.1 COVID (2020–2021):

The GPR coefficient is negative and statistically significant in both sectors (Energy: −0.0029, $t = -5.22$, $p < 0.01$; A&D: −0.0009, $t = -2.60$, $p < 0.01$). Both magnitudes exceed those of the full OOS sample, indicating the decorrelation mechanism was at its strongest during the pandemic. During systemic shocks, sector-specific drivers - collapsing oil demand for Energy, continuity of defence procurement for A&D - dominate over market-wide co-movement.

5.6.2 Ukraine and the energy crisis era (2022):

Energy retains a negative and significant GPR coefficient (−0.0006, $t = -3.21$, $p < 0.01$), though smaller in magnitude relative to COVID. The A&D coefficient is not statistically significant.

The persistence of the Energy result is intuitive: the 2022 episode was a direct supply shock to oil and gas markets, exactly the kind of disruption Kilian (2009) identifies as driving sector-specific responses through the supply channel. The fading of the A&D effect suggests the haven role of defence equities was already weakening by 2022, prefiguring the structural shift documented next.

5.6.3 Post-Ukraine era (2023-2025) :

The Energy GPR coefficient is no longer statistically significant, while the A&D coefficient flips sign from negative to positive (+0.0003, $p = 0.054$). The sign change is marginal at conventional thresholds but economically meaningful: A&D moves from behaving as a partial haven, with decorrelation under stress, to behaving as a crisis amplifier, with stronger co-movement during elevated GPR.

The plausible mechanism is that markets have structurally re-priced the sector following the Ukraine war. Before 2022, geopolitical events were typically transient, and defence equities attracted short-term safe-haven flows; the Ukraine war demonstrated that sustained conflict translates into multi-year defence procurement commitments and rising NATO budgets (SIPRI, 2024). The shift from short-term hedge to procyclical exposure is consistent with the Efficient Market Hypothesis: markets efficiently incorporate the new reality that sustained conflict produces persistent defence revenue, rather than treating it as a transient shock. While initiating this study, the working hypothesis was that defence equities act as a safe investment during war; the empirical evidence instead shows that the haven property holds only when conflict is expected to be temporary. The wider pattern across the three subsamples - strongest decorrelation during COVID, fading during Ukraine, sign-flipped post-Ukraine - points to market learning over time: as investors anticipate that geopolitical risk will persist rather than resolve quickly, they price the sectors as ordinary cyclical exposures rather than havens. The

result extends the sectoral asymmetry documented by Gheorghe and Panazan (2025) by adding a temporal dimension, and is consistent with the conditional CAPM logic of Lewellen and Nagel (2006).

5.7. Seismograph alarm validation:

The first research question asks whether the seismograph can detect regime shifts in real time rather than only after the fact. Across 2020–2025, the seismograph fires 111 times for Energy and 125 times for A&D. The higher A&D count is consistent with the sector's documented sensitivity to geopolitical news flow (Apergis et al., 2017; Gheorghe and Panazan, 2025), and the comparable order of magnitude suggests that the two-standard-deviation threshold produces a usable alarm density.

Event Date	Event	Nearest alarm	Lead/Lag (days)	Nearest alarm	Lead/Lag (days)
11-03-2020	WHO declares COVID-19 pandemic	05-03-2020	-6	03-03-2020	-8
16-03-2020	US market COVID crash (circuit breaker)	05-03-2020	-11	03-03-2020	-13
24-02-2022	Russian invasion of Ukraine begins	24-11-2021	-92	09-11-2021	-107
26-09-2022	Nord Stream pipeline sabotage	14-09-2022	-12	19-10-2022	23
07-10-2023	Hamas attack on Israel (calendar; nearest TD used)	30-08-2023	-38	09-10-2023	2

Table 5.7: Seismograph alarm validation against major events

Table 5.7 reports the lead or lag of the nearest sector-specific alarm relative to five anchor events. The seismograph alarmed six days before the WHO pandemic declaration, eleven days before the March 2020 market crash, ninety-two days before the Russian invasion (almost certainly capturing the troop buildup phase that financial markets were tracking in late 2021), and twelve days before the Nord Stream incident. The Hamas attack is the one event the seismograph did not lead, with the A&D alarm registering two days after. Three of the five events were therefore detected with a lead time of at least a week. The consistent direction of the result - leading rather than lagging on the majority of major shocks - supports the

seismograph framework of Hooper (2025) as an operational early-warning system rather than a purely descriptive one.

5.8. Forecast accuracy and robustness checks:

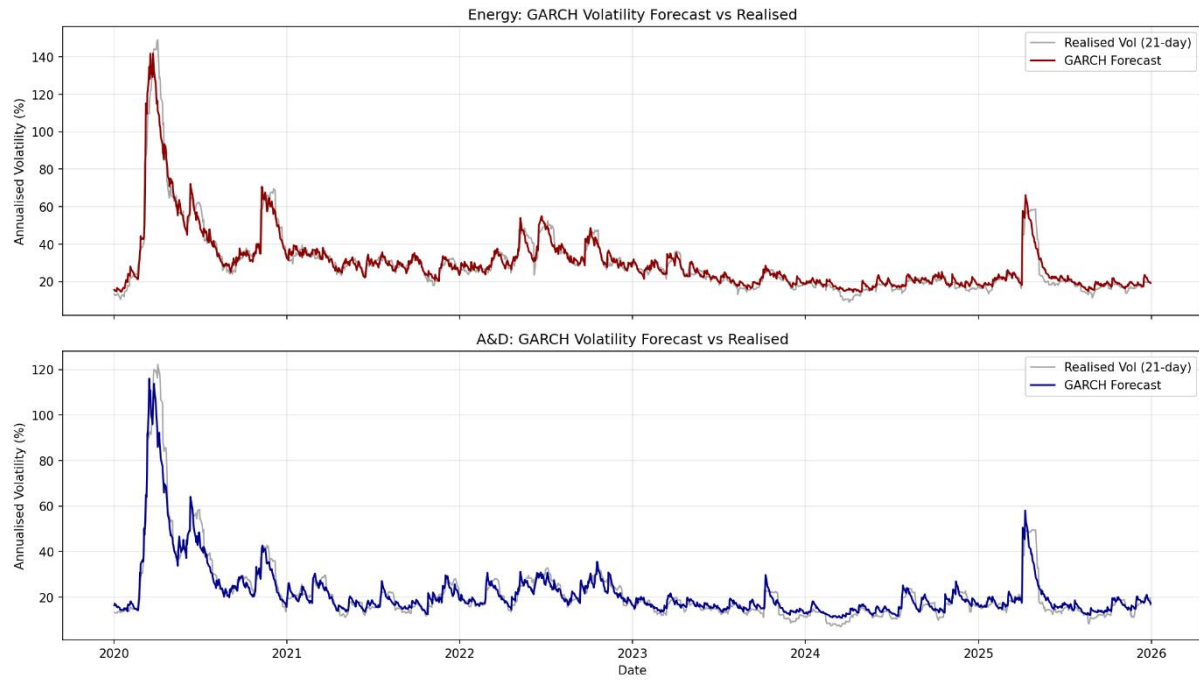


Figure 5.3: Volatility Forecasted vs Realised

The seismograph's timing alarms are paired with GARCH-based duration forecasts as the second layer of the early-warning system. GARCH-implied conditional volatility for each sector is compared to a 21-day rolling realised volatility (Table 5.8). The mean absolute error is 9.36% for Energy and 11.97% for A&D, both meaningfully lower than the 17.73% MAE for the S&P 500 itself, and both below the 20% threshold conventionally used as the benchmark for acceptable accuracy (Bouras et al., 2019; Adekoya et al., 2025). This supports the duration layer of the seismograph framework.

Series	MAE	RMSE	MAE %	RMSE %
Energy	2.78	4.45	9.36	14.99
A&D	2.6	4.12	11.97	18.97
Market (S&P 500)	3.08	5.19	17.73	29.89

Table 5.8: GARCH forecast accuracy: MAE and RMSE

A variance ratio test on the dynamic beta series addresses whether the OOS regime genuinely differs from training. The OOS variance of Energy's dynamic beta is 2.47 times its training-period variance ($p < 0.001$), and A&D's OOS variance is 1.48 times its training-period variance ($p < 0.001$). The variance ratios are economically large and statistically unambiguous: the dynamic beta series is structurally more dispersed during the OOS period, validating both the seismograph's increased alarm frequency in 2020–2025 and the regime-dependent significance of GPR documented in Section 5.3.

A final robustness check adds a VIX–GPR interaction term to the OOS regression. The interaction is not statistically significant for either sector, meaning GPR's effect on dynamic beta does not depend on the VIX regime. The decorrelation mechanism in Section 5.4 therefore operates independently of the VIX regime, indicating a structural rather than fear-driven effect.

Series	Train Var	OOS Var	Ratio	F-statistic	p-value
Energy Beta	0.0793	0.1955	2.47	2.47	<0.001
A&D Beta	0.056	0.0826	1.48	1.48	<0.001

Notes: Variance ratio of OOS to training-period dynamic beta variance, with F-test of equal variances (training: 1,258 obs; OOS: 1,507 obs)

Table 5.9: Variance Ratio Test

Variable	Energy sector			A&D sector		
	Coef	t-stat	p-value	Coef	t-stat	p-value
GPRD Index	−0.0025***	−14.89	<0.001	−0.0012***	−10.80	<0.001
Log VIX	1.4808***	4.27	<0.001	0.6931***	2.98	0.003
GPR × VIX	−0.0036	−1.63	0.103	0.0002	0.15	0.881
Log Total Returns	5.3194***	4.57	<0.001	3.5932***	4.61	<0.001
R ²	0.1457			0.0933		

*Notes: Out-of-sample period: 1,507 observations. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table 5.10: VIX–GPR interaction regression (OOS period)

5.9. Implications for portfolio managers and risk practitioners:

Three operational implications follow from the empirical findings. First, the lag-10 result suggests GPR observed today is informative about sector beta approximately two trading weeks into the future, long enough to implement defensive hedges through index options, sector ETF rebalancing, or reduction in market exposure, but short enough that the signal does not decay before action can be taken. Combined with the seismograph's 6–12 days of lead time on the major events of the OOS period, this gives institutional portfolio managers a two-part early warning system: GPR for forward-looking timing, and the seismograph for real-time confirmation.

Second, dynamic beta provides a regime-aware complement to conventional value-at-risk models that rely on static correlations. VaR models tend to break down precisely when they are needed most - during crisis periods when correlations spike and the historical distribution becomes uninformative (Lewellen and Nagel, 2006). The dynamic beta framework detects regime shifts as they occur, allowing risk managers to flag elevated risk before conventional models catch up.

Third, the level-versus-shock channel distinction implies different trading strategies for different types of geopolitical environments. Sustained tension regimes call for rotation into decorrelating sectors that mechanically offer flight-to-quality benefits; sudden-shock days call for short-term volatility hedging followed by mean-reversion positioning. The findings in 5.6 add a regime-conditional caveat: the A&D haven property is not stable, and any strategy relying on it as a permanent hedge would have broken down between 2022 and 2023.

This study contributes along three axes. First, it moves Hooper's (2025) seismograph framework from concept to operational tool by quantifying a ten-trading-day predictive horizon. Second, it adds a temporal dimension to the sectoral heterogeneity documented by

Apergis et al. (2017) and Gheorghe and Panazan (2025), showing the GPR–beta relationship is itself regime-dependent. Third, it supports the conditional CAPM view (Fabozzi and Francis, 1978; Lewellen and Nagel, 2006) by demonstrating that the relationship between beta and its underlying drivers varies across regimes, not only between calm and stressed conditions.

The findings are directly relevant to macro hedge funds, pension funds with large sectoral exposures, and bank risk management desks. Any institution holding positions in geopolitically exposed sectors where early detection of regime shifts has operational value.

6. LIMITATIONS:

The GPR index is constructed from English-language press (Caldara and Iacoviello, 2022), which may underrepresent geopolitical events that are salient in non-Western media. As the index remains the daily-frequency standard in the literature, this is a measurement constraint shared across the field rather than one specific to this study.

The analysis is restricted to US sectors, so the flight-to-quality and decorrelation results may be partly US-specific, given the dollar's role in commodity pricing and US dominance of global defence procurement (SIPRI, 2024). Multi-country replication is proposed in Section 7.

The five-year training window is short, and the largest seismograph alarm clusters occurred around the 2020 pandemic. As shown in Section 5.2, this reflects the framework's sensitivity to sector-versus-market volatility differentials rather than a purely geopolitical signal; the geopolitical contribution is instead identified through the regression evidence in Sections 5.5 and 5.6.

GARCH parameters are estimated on the full sample rather than re-estimated through the OOS window, a mild form of look-ahead bias. GARCH(1,1) parameters are typically stable across sub-windows of equity returns (Hansen and Lunde, 2005), so the qualitative findings are

unlikely to change under a stricter specification; rolling-window GARCH is proposed in Section 7.

The dynamic beta formula holds ρ fixed at its training-period value, which means the decorrelation result in Section 5.4 likely understates the true correlation collapse under stress rather than overstating it. The Dynamic Conditional Correlation framework of Engle (2002) would relax this and is proposed as the methodological extension in Section 7.

Explanatory power is modest, with R^2 ranging from 0.03 in training to 0.20 in the lagged specification. This is expected for a framework built to detect regime shifts rather than fully explain beta variation; the significance of the GPR coefficient is independent of overall fit.

The lead times in Sections 5.7 and 5.8 are descriptive, not causally identified. Alarms cluster around major events, but a formal identification strategy is needed to rule out anticipatory repositioning or a common unobserved driver; event-study windows are proposed in Section 7.

The Ukraine subsample (251 observations) limits statistical power, and the post-Ukraine A&D result is only marginally significant ($p = 0.054$). It is therefore presented as suggestive evidence of a structural shift rather than a confirmed finding, and would benefit from a longer post-Ukraine sample as more data becomes available.

7. CONCLUSIONS:

The first research question asked whether the seismograph framework could detect regime shifts in real time rather than only after the fact. The empirical evidence supports an affirmative answer. Across the OOS period (2020–2025), the seismograph fired 111 times for Energy and 125 times for A&D, with alarm dates clustering around the five major geopolitical events of the period. Of these five events, three were detected with at least a week of lead time (WHO COVID-19 declaration: -6 days; March 2020 market crash: -11 days; Nord Stream sabotage: -

2 days), one was detected with several weeks of lead time (Russian invasion: -92 days), and one was detected with a two-day lag (Hamas attack). The framework therefore operates as a real-time early-warning system in the majority of cases, validating Hooper's (2025) conceptual proposal through systematic empirical testing.

The second research question asked whether GPR carries information about future sector beta, and over what horizon. The lagged-GPR specification documents a clear pattern: in the OOS period, GPR at all four tested horizons (lag 0, 1, 5, 10) is statistically significant at the 1 percent level, with the lag-10 coefficient carrying the largest magnitude in both sectors (Energy: -0.0011; A&D: -0.0009). GPR observed today is therefore informative about sector beta approximately two trading weeks into the future, supporting the seismograph framework's role as a forecasting tool rather than a purely descriptive one. Although the contrast with the training period, where none of the lags reach significance at conventional thresholds, confirms that this predictive content emerges only during periods of elevated geopolitical activity.

The third research question asked whether the GPR–beta relationship is uniform across crisis regimes or varies across them. The subsample regressions demonstrate clear regime-dependence: the decorrelation mechanism was strongest during COVID (2020–2021), started to weaken during Ukraine (2022), and was absent or reversed during the post-Ukraine era (2023–2025). The most striking result is the sign change in the A&D coefficient between the Ukraine and post-Ukraine subsamples, moving from non-significant to positive and marginally significant as a structural re-pricing of the sector following the demonstrated transition from transient geopolitical shocks to sustained military procurement environments. This pattern is consistent with the Efficient Market Hypothesis, as we earlier discussed that the markets efficiently incorporate the new reality.

The empirical contribution is three-fold: documentation of a regime-dependent GPR–beta relationship that switches on out-of-sample but is absent in training; identification of a ten-day predictive horizon from GPR to sector beta; and documentation of a post-Ukraine structural shift in A&D from haven to amplifier.

The methodological contribution is the operationalisation and out-of-sample validation of Hooper's (2025) seismograph and compass framework. Where Hooper proposes the architecture conceptually, this study implements it on two specific sectors, tests it against five major geopolitical events, and quantifies its predictive lead times, extended with a GARCH-based duration forecast.

The practical contribution is the demonstration that geopolitical risk can be incorporated into operational portfolio risk management through a quantifiable lead-time signal. The combination of a ten-day GPR predictive horizon and a real-time seismograph alarm provides institutional portfolio managers with a two-part early warning system suitable for hedging timing decisions, sector rotation strategies, and complementary risk monitoring alongside conventional value-at-risk models.

Future work should extend the framework along five dimensions: multi-country application (UK, Eurozone, Japan); sector-specific GPR sub-indices via keyword filtering; the Dynamic Conditional Correlation specification of Engle (2002) to allow time-varying ρ ; event-study windows around twenty or more specific geopolitical events to strengthen causal interpretation; and rolling-window GARCH re-estimation to address the look-ahead concern. Also, extending the sample to include the 2007-2009 financial crisis to test whether the framework correctly signals the GFC would be a significant finding, as this would mean that this framework not only applies to geopolitical events but also to non-geopolitical events, too, and can be improved to capture those regime shifts.

Geopolitical risk has become a structural feature of contemporary financial markets, not an occasional disturbance. The framework developed here is intended not only for institutional risk management but also as a tool for broader societal understanding of how geopolitical events are priced into financial markets, supporting preventative measures during periods of uncertainty. Frameworks that detect when sector-market relationships break down, rather than averaging across calm and stressed regimes, are increasingly relevant for both portfolio risk management and academic research on uncertainty pricing.

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APPENDIX NOTE:

The full Python code used to implement the empirical work described in Section 4 is available upon request. The code consists of 936 lines across 25 sequential steps, covering data loading, diagnostic testing, GARCH estimation, dynamic beta construction, seismograph alarm generation, out-of-sample regression, subsample analysis, and robustness checks. The file is named `Research_Project_FINAL.py`